

Review retrospective:
*the quartic–sextic closed-form core — a dead open and an
ill-typed lambda*

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Abstract

A soundness review of `Laplace/TwoD/QuarticSextic.lean`: the closed-form core that specialises the generic additively-separable factorisation theorems to the (quartic, sextic) pair, $L(x, y) = x^4/24 + y^6/720$, and that the quartic–sextic asymptotic-packaging files consume. Result: **a clean fidelity pass with two edits** — a dead open `Real` and a docstring that wrote a two-variable potential as a one-variable lambda.

1 Setting

This file is the bridge between the reviewed generic root (`AddSeparable`) and the $(2, 3)$ asymptotic specialisations (partition, moment): it discharges the integrability side-conditions for the quartic and sextic Boltzmann weights and instantiates the three generic theorems, then substitutes the 1D closed forms to give the explicit Z_{2D} value.

2 What was reviewed

Two private integrability lemmas plus five public theorems: the partition factorisation $Z_{2D} = Z_q Z_s$, its closed-form value $\frac{1}{2}(24/t)^{1/4}\Gamma(1/4) \cdot \frac{1}{3}(720/t)^{1/6}\Gamma(1/6)$, strict positivity, the separable-observable moment factorisation, and the mixed-coordinate covariance vanishing.

3 Fidelity / soundness findings

Sound; no theorem-level mismatch. The closed form matches the product of the 1D `quartic_partition/sextic_partition` values, positivity follows from the 1D factors, and the factorisation/covariance theorems are the expected specialisations of the reviewed generic add-separable theorems, with the right nonzero-partition and integrability hypotheses. **One docstring inaccuracy, fixed.** The `partitionFunction_quarticSextic_factor` docstring described the potential as “ $x \mapsto x^4/24 + y^6/720$ ” — an ill-typed one-variable lambda whose body mentions y . Corrected to $(x, y) \mapsto x^4/24 + y^6/720$ (comment-only).

4 Simplifications applied

One dead open. `open Real` is dead (`Real.exp/Real.Gamma` qualified, bare $(24/t)^{(\text{real})}$ is `rpow`); `open MeasureTheory` stays — bare `Integrable` appears in the two private lemmas and the `hf/hg`

hypotheses. Module green at 2701 jobs, no warnings, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None. The reviewer’s proposal to inline the single-use `have` in the two private lemmas (and drop a likely-redundant `one_mul` from the `simp` set) was declined as proof-golf.

6 Lessons

- **Ill-typed informal notation is a fidelity nit worth fixing.** “ $x \mapsto x^4/24 + y^6/720$ ” parses (to a reader) as a one-argument function, but the body is binary; it is the documentation analogue of a type error. Cheap to fix, and it removes a genuine point of confusion about the potential’s arity.
- **The closed-form core is where the explicit constants enter.** Reviewing it after the asymptotic packagers (which only repackage) confirmed that the 24, 720, $\Gamma(1/4)$, $\Gamma(1/6)$ all trace to the 1D `quartic_partition/sextic_partition` — closing the provenance of every constant the $(2, 3)$ arc uses.

7 Follow-ups

- The two remaining `QuarticSextic*` closed-form files, `QuarticSexticMoments` and `QuarticSexticOddMoments`.
- Proof-golf forward tide: inline the single-use `haves` in the private integrability lemmas (here and in the `kthkth` siblings).